

Socioeconomic Impact Evaluation of Terrestrial Protected Areas in Madagascar based on large national surveys

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Abstract Protected Areas are the most widely used tool for biodiversity conservation. However, their implementation raises concerns about the well-being of local populations, especially when they are very poor and dependent on natural resources, as is the case in Madagascar. We are using geolocated socioeconomic surveys spanning a period of protected area creation (2008-2021), and extending the analysis to the previous period (1997-2008) in order to verify the credibility of the comparison. Socio-environmental indicators derived from spatial data are used to match rural areas with similar probabilities of being affected by the creation of protected areas, before estimating the effects using an approach combining matching and difference-in-difference. The analysis, conducted in accordance with a pre-analysis plan submitted prior to any analysis,

shows that our empirical model would have detected average effects of at least 7.5 percentiles of the wealth index for rural households. The results indicate that, over the period considered, the creation of protected areas had no statistically significant average effect on the well-being of households located within 10 km of protected areas created after 2008, as measured by the intra-cluster dispersion of this index. Robustness test conducted using other survey sources support this conclusion. Further work is needed to determine whether this lack of effect reflects a trade-off between positive and negative impacts, high heterogeneity of effects, or limited implementation of conservation measures in a context of weak governance and underfunding of protected areas.

1 Results

1.1 Overall impact on livelihoods

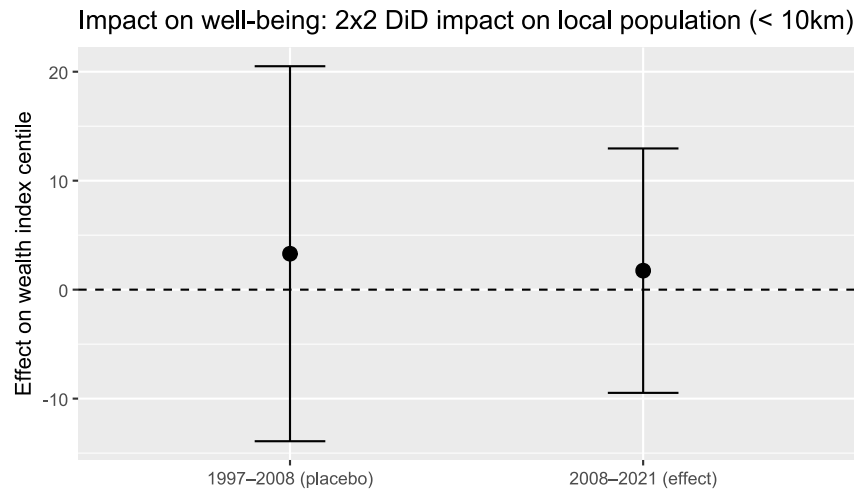


Figure 1: Impact on well-being: 2×2 difference-in-differences estimates

1.2 Effect on inequalities

1.3 Heterogeneity